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Debt Contracting on Management

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1 Big picture

- **Question.** Why do lenders write *change-of-management restrictions* (CMRs) into private loan contracts, and what happens to CEO turnover and firm performance when such clauses are in force?
- **What a CMR is.** A CMR is a covenant that makes a change in management — in practice, usually a change in the CEO — an event of default or a trigger for lender approval. Instead of contracting on financial ratios or specific actions, the lender contracts directly on *who runs the firm*.
- **Setting.** The paper hand-collects CMR clauses from EDGAR loan agreements matched to DealScan, Compustat, CRSP, and ExecuComp. The core sample contains **15,501 private loan contracts** for **4,411 borrowing firms**. The analysis is mostly at the **loan-package level**.
- **Why the paper matters.** Hart and Moore (1994) argue that debt contracts must respond to the fact that human capital is inalienable. This paper shows that lenders do not just adjust maturity or collateral — they sometimes write contracts directly on **managerial retention and succession**. That makes human capital risk a first-order debt contracting issue.
- **Main competing hypotheses.**
 - **Risk hypothesis:** lenders use CMRs to reduce risks from CEO turnover, especially loss of firm-specific human capital, replacement uncertainty, and talent-retention frictions.
 - **Collusion hypothesis:** lenders and CEOs use CMRs to entrench the CEO, either to preserve the lending relationship or in exchange for a higher interest rate.
- **Main descriptive fact.** CMRs are uncommon overall but concentrated in settings where managerial human capital appears especially important. At the **contract level**, only **3.4%** of loans contain a CMR. At the **firm level**, **8.5%** of firms have one at some point, and among small firms the incidence rises to **17.2%**.
- **Main baseline result.** CMRs are more likely in loans to **smaller, riskier, less tangible** borrowers, in more tightly monitored contracts, and in loans with shorter maturity and collateral. The pattern is exactly what one would expect if lenders worry that losing the CEO could materially damage repayment prospects.
- **Main human-capital result.** CMRs are more likely when the CEO is the **founder**, when the industry tends to promote CEOs **from within**, and when the firm has **no obvious heir apparent**. These are all settings in which replacement is harder or the manager's skills are more firm-specific.
- **Main contracting-frictions result.** CMRs are also more likely when the firm has trouble retaining a valuable CEO: when the CEO owns more stock, has no unvested equity to lose, operates in a state with **weak noncompete enforcement**, or is close to **retirement age**.
- **Main pricing result.** After correcting for the simultaneity of covenant design and loan pricing, the paper finds that CMR inclusion is associated with a **lower predicted loan yield**. That

is inconsistent with the collusion view and much more consistent with efficient contracting that protects lenders.

- **Main turnover result.** When a CMR is binding, the likelihood of CEO turnover falls by roughly **5.2 percentage points**, which is about a **55% decline** relative to the unconditional turnover rate in the turnover sample.
- **Main performance result.** In the full sample, CMRs are not linked to a strong unconditional improvement in firm value or operating performance. But in the subsamples where CMRs should matter most — firms facing retention frictions or firms closer to default — future **Tobin's Q** and **operating cash flow** improve.
- **Bottom line.** Lenders use CMRs as targeted control rights over management when managerial human capital is especially important for debt value. CMRs appear to be a retention and succession device, not an entrenchment favor for CEOs.

2 Data and measurement (institutional setting and key objects)

2.1 Why private loan contracts are the right laboratory

- The paper studies **private bank loans**, not public bonds. This is useful because private contracts contain richer covenant language and private lenders are typically stronger monitors.
- The contracts are available through **EDGAR**, which lets the authors read the actual legal text rather than infer control rights indirectly from outcomes.
- The institutional environment is especially appropriate because banks can write detailed contract terms, monitor borrowers closely, and renegotiate if needed.

Interpretation. If one wants to know whether debt contracts ever speak directly to management retention, syndicated loan agreements are the natural place to look.

2.2 Sample construction

- Start with DealScan package-level loans matched to Compustat and CRSP.
- Keep nonfinancial, nonregulated firms and require loans originating after **January 1, 1995**, when EDGAR contract availability becomes reliable.
- This yields **27,037** unique loan packages from **6,046** firms with valid CIK identifiers.
- The authors then search firm filings (10-K, 10-Q, and 8-K) for credit agreements, amendments, and restatements using regular-expression text searches.
- The text search produces **59,719** candidate filings. Matching these to DealScan based on firm identity and dates yields the final contract sample of **15,501** loans from **4,411** firms.

Interpretation. The paper’s contribution is not just an empirical regression design; it is also a major data-construction exercise that makes these covenants observable.

2.3 How the paper identifies CMR clauses

- The authors first run a broad textual search for change-related and management-related language.
- That search finds **16,207 paragraphs** from **7,155 loan packages** with possible change-of-management wording.
- Manual filtering narrows this to about **2,100** paragraphs and ultimately to **565** candidate contracts with a true CMR.
- After dropping mismatched deals, the final CMR sample contains **532 CMR contracts** from **376 firms**.
- In **96%** of CMR contracts, the clause explicitly references the **CEO**. The wording usually covers management departures broadly, rather than only board-initiated dismissals.

Interpretation. CMRs are rare enough that they have to be hand-collected, but concrete enough that the paper can distinguish real governance rights from generic boilerplate.

2.4 How restrictive a CMR can be

- CMR clauses vary in legal strength.
- At the strict end, *any* change in management can trigger default automatically.
- Intermediate versions trigger default for voluntary departures, or for changes without prior lender consent.
- At the least restrictive end, the borrower must obtain *ex post* lender approval for a management change.

Interpretation. The paper is not only about whether a clause exists; it also recognizes that lenders can choose stronger or weaker control rights.

2.5 Main explanatory variables

- **Borrower risk and scale:** $\log(AT)$, modified Altman’s *Z*-score, tangibility, market-to-book, operating cash flow, leverage, and whether the firm is rated.
- **Human capital proxies:**
 - **Founder CEO:** indicator that the CEO in place at loan initiation is effectively the founder.
 - **% Insider (Ind.):** industry-level share of incoming CEOs promoted from within, from Cremers and Grinstein (2014).

- **No Heir Apparent:** no top executive other than the CEO holds the title president or COO.
- **Contracting-frictions proxies:**
 - **CEO Ownership %**
 - **CEO No Unvested Equity**
 - **Low NC Enforcement:** headquartered in a state with below-median noncompete enforceability.
 - **CEO Retirement Age:** CEO near the retirement discontinuity.
- **Loan and syndicate controls:** spread, number of lenders, scaled loan amount, maturity, collateral, financial covenants, performance pricing, lead share retained, and whether the lead is local.

Interpretation. The explanatory variables map tightly into the theory: borrower risk, uniqueness of the manager, ease of replacement, and difficulty of retention.

2.6 Main outcomes

- **Loan pricing:** All-In Spread Drawn, modeled through a selection-adjusted counterfactual yield framework.
- **CEO turnover:** indicator for CEO turnover in ExecuComp firm-years, with active-CMR timing based on facility start and end dates.
- **Future firm value:** changes in Tobin’s Q.
- **Future operating performance:** changes in operating cash flow scaled by assets.

Interpretation. The paper does not stop at “who gets a CMR?” It also asks whether the clause changes financing terms, managerial turnover, and eventual firm outcomes.

2.7 Descriptive facts from Table I

- CMR loans are issued to much **smaller** firms: mean $\log(AT)$ is **5.17** versus **6.82** in non-CMR loans.
- They are issued to **riskier** firms: mean Z -score is **0.77** versus **1.43**.
- Founder CEOs are much more common in CMR loans: **45%** versus **20%**.
- CEO ownership is higher: **4%** versus **2%**.
- CMR loans have larger spreads (about **245 bps** versus **202 bps**), fewer lenders (**3.4** versus **7.7**), shorter maturity (**34.5** versus **45.3** months), and are more likely to be collateralized (**74%** versus **52%**).

Interpretation. Even before multivariate analysis, the raw data already look like a story about fragile borrowers with valuable managers and close lender monitoring.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: risk versus collusion

The paper organizes its empirical tests around two competing views.

- **Risk hypothesis.** A CEO departure can hurt debt value through operating disruption, loss of firm-specific human capital, and uncertainty about the replacement. If shareholders cannot perfectly retain the CEO, lenders may want direct contractual control.
- **Collusion hypothesis.** The CEO may prefer a covenant that makes dismissal costly. Lenders might agree either to protect the relationship or in exchange for a higher rate.

Interpretation. The paper’s structure is elegant: first explain where CMRs show up; then ask whether the observed pattern looks like risk management or CEO entrenchment.

3.2 Baseline CMR-inclusion specification

The baseline empirical model is a loan-level probit for whether a package contains a CMR:

$$CMR_i^* = \alpha + X_i' \beta + \varepsilon_i, \quad CMR_i = 1\{CMR_i^* > 0\}. \quad (1)$$

Here X_i includes borrower characteristics, loan characteristics, syndicate characteristics, macro controls, year fixed effects, and industry fixed effects.

In practice, the baseline specification is built around variables such as size, Z -score, tangibility, market-to-book, operating cash flow, leverage, rating status, log number of lenders, maturity, collateral, number of financial covenants, and performance pricing.

Interpretation. This regression answers the first-order question: after controlling for ordinary loan and borrower features, which environments make lenders want explicit control over management?

3.3 Human-capital and contracting-frictions extensions

The paper then augments the baseline probit with specific proxies for the importance of managerial human capital and the difficulty of retaining talent:

$$CMR_i^* = \alpha + \beta HC_i + \gamma Friction_i + \lambda(HC_i \times Friction_i) + Controls_i + \varepsilon_i. \quad (2)$$

Where:

- HC_i is proxied by **Founder CEO**, **% Insider (Ind.)**, and **No Heir Apparent**.
- $Friction_i$ is proxied by **CEO Ownership %**, **CEO No Unvested Equity**, **Low NC Enforcement**, and **CEO Retirement Age**.

Interpretation. These extensions let the authors go beyond “small and risky firms get CMRs” and show that the clauses appear exactly where management is harder to replace and harder to keep.

3.4 Loan-pricing test for the collusion hypothesis

To separate efficient contracting from collusion, the paper uses the Lee (1978) / Heckman-style simultaneous design also used by Miller and Reisel (2012) and Bradley and Roberts (2015). The procedure has three steps.

Step 1: selection equation

$$CMR_i^* = \alpha + X_i'\theta + Z_i'\xi + \zeta_i. \quad (3)$$

Step 2: pricing equations conditional on CMR status

$$\log Yield_i^{NoCMR} = X_i^{NoCMR}\beta^{NoCMR} + IMR_i^{NoCMR} + \nu_i^{NoCMR}, \quad (4)$$

$$\log Yield_i^{CMR} = X_i^{CMR}\beta^{CMR} + IMR_i^{CMR} + \nu_i^{CMR}. \quad (5)$$

Step 3: does CMR inclusion line up with a yield discount?

$$CMR_i^* = \alpha + \delta \left(\log \widehat{Yield}_i^{NoCMR} - \log \widehat{Yield}_i^{CMR} \right) + Z_i'\beta_c + \varepsilon_i. \quad (6)$$

The key parameter is δ . If the risk hypothesis is right, a CMR should reduce expected loan yields, so larger predicted savings should be associated with more CMR use: $\delta > 0$. Under collusion, the relation should be null or negative.

Interpretation. This is the paper’s cleanest “horse race” between the two stories, because the cross-sectional covariates alone cannot fully distinguish value-preserving retention from CEO entrenchment.

3.5 CEO-turnover specification

To test whether CMRs are actually binding, the paper estimates a turnover model of the form

$$Turnover_{it} = 1\{\alpha + \beta CMRBinding_{it} + \phi Performance_{i,t-1} + \psi CEOControls_{i,t-1} + YearFE_t + u_{it} > 0\}. \quad (7)$$

Key controls include one-year or three-year total shareholder return, lagged ROA, high CEO ownership, CEO retirement age, and CEO tenure. The sample is restricted to ongoing firms so the model captures intentional turnover decisions rather than mergers or firm failure.

Interpretation. If firms and boards truly respect the covenant, turnover should fall while the CMR is active and rise again when the clause is no longer binding.

3.6 Future-performance specification

Following Nini, Smith, and Sufi (2009), the paper studies changes in future firm value and operating performance around CMR implementation:

$$Performance_{i,t} = \alpha + \beta CMR_{i,t-1} + \delta Performance_{i,t-1} + \gamma_t + \varepsilon_{i,t}, \quad (8)$$

where the dependent variable is the difference between the two-year average after contract initiation and the two-year average before contract initiation.

The authors estimate this both unconditionally and after decomposing the CMR effect across firms with stronger retention frictions or weaker financial health.

Interpretation. These regressions ask whether CMRs merely lock in managers, or whether doing so creates value in the settings where lenders are most worried about losing them.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the determinant regressions are informative

- The baseline CMR regressions include rich borrower, loan, syndicate, macro, year, and industry controls.
- The analysis is done at the package level, the level at which covenants are actually set.
- The strongest interpretation does not come from one coefficient alone, but from the **pattern** across many coefficients: small firms, risky firms, founder CEOs, internal-promotion industries, absent heirs apparent, weak noncompete enforcement, and retirement-age CEOs.

Why this matters. A single proxy could always be noisy. The paper is persuasive because many different proxies all point in the same direction.

4.2 How the paper distinguishes risk management from collusion

Pure cross-sectional correlations cannot cleanly separate risk from CEO entrenchment, because a powerful founder may be both hard to replace and hard to fire. The pricing test is therefore crucial.

- The paper models covenant choice and pricing jointly.
- It uses syndicate characteristics — such as number of lenders, lead allocation, and local lead — as exclusion restrictions in the covenant-cost equation.
- The logic is that in a competitive lending market, these syndicate features affect the cost or ease of writing a covenant but should not directly determine loan yields except through covenant structure.

Why this matters. The positive estimated relation between CMR use and the counterfactual yield saving is the cleanest evidence against the collusion story.

4.3 Why the turnover evidence is stronger than a static cross-section

- The paper does not merely compare firms that ever have a CMR to firms that never do.
- It constructs an indicator for when the CMR is actually **binding** based on active loan dates.
- It controls for past performance and CEO characteristics.
- It adds a **CMR Firm** indicator in some specifications; that variable is insignificant, while the active-binding indicator remains strongly negative.
- Figure 1 shows abnormal turnover falling when the CMR starts and rising again when it ends.

Why this matters. That timing pattern makes a simple omitted firm-type story much less plausible.

4.4 Supportive mechanism evidence

Several additional tests reinforce the human-capital interpretation.

- **Key man insurance:** CMRs are less likely when a borrower has key-man insurance, and less likely when the insurance amount is larger. That is what one would expect if both insurance and CMRs address human-capital risk.
- **Restrictiveness:** more restrictive CMRs are more likely when human-capital risk is higher, especially for founder CEOs and industries with more inside promotion.
- **Covenant violations:** the usual positive relation between covenant violation and CEO turnover is weaker when a CMR is in place.
- **Legal evidence:** in *State National Bank v. Farah Manufacturing Company*, lenders explicitly used a CMR to influence CEO selection.

Why this matters. The paper does not rely on one empirical margin only; it triangulates the mechanism from pricing, turnover, legal narrative, and other contract substitutes.

4.5 Alternative explanations the paper works to rule out

The paper explicitly considers several rival stories.

- **Debt overhang:** little support; long-term debt maturity, R&D intensity, and VIX do not line up with the CMR story in the way overhang would predict.
- **Large blockholders:** the share held by the largest block is not positively related to CMR use.
- **Takeover deterrence:** takeover-threat proxies are insignificantly negative, not positive.

- **Bundling with other contract terms:** results survive controls for change-in-control clauses and a wide set of other covenant features.
- **Lead-bank fixed taste for CMRs:** most lead banks in the CMR sample also lead non-CMR loans, making a pure lender-style explanation less convincing.

Interpretation. These tests do not make the design fully experimental, but they reduce the space for obvious confounds.

4.6 What the paper can and cannot claim

- The paper is strongest on documenting that CMRs exist, on showing where they are used, and on showing that they appear to affect turnover.
- The future-performance tests are more **suggestive** than fully causal, because firms are not randomly assigned CMRs.
- The pricing design helps with simultaneity, but exclusion restrictions for covenant-cost versus yield are still an identifying assumption.

Bottom line. The evidence is highly consistent with the risk hypothesis, but the paper should still be read as very strong empirical support rather than a literal randomized causal design.

5 Tables and figures

5.1 Table I: firm, CEO, and loan characteristics

Table I
Firm, CEO, and Loan Characteristics

Panel A: Summary Statistics

	N	Mean	Median	SD
Firm characteristics				
CMR Proxy	15,501	0.00	0.00	0.20
Log(S)	14,260	4.79	4.77	1.77
Size	14,267	1.41	1.48	1.34
Tangibility	14,269	0.20	0.22	0.40
ROE	14,331	2.47	1.83	3.00
Operating CF	14,269	0.16	0.13	0.14
Leverage (Debt)	14,269	0.16	0.13	0.14
Ratio	15,501	0.45	0.40	0.20
Term Spread	15,501	1.15	1.21	0.97
Credit Spread	15,501	0.86	0.86	0.24
Ratio	15,501	1.00	1.00	0.00
Loan characteristics				
Log(S)	15,452	4.80	4.79	1.76
% Senior Debt	15,452	0.07	0.00	0.25
Transfer CEO	7,489	0.00	0.00	0.40
CEO Ownership %	7,489	0.02	0.00	0.04
CEO Retirement Age	7,254	0.00	0.00	0.27
No. New Appoint	7,254	0.00	0.00	0.46
CEO Age	7,254	50.10	50.00	4.57
CEO Tenure	7,254	6.10	6.00	2.20
No. New CEO	7,254	0.00	0.00	0.46
Loan characteristics				
CMR Proxy	15,501	0.00	0.00	0.18
All-in Spread (bps)	15,501	103.19	173.06	104.48
Number of Lenders	15,488	7.51	1.00	7.09

Table I - Continued						
Panel B: Univariate Analysis						
	Loan Contract with a CMR		Loan Contract with no CMR		Difference in Means	Wilcoxon Rank-Sum Test
	Mean	Median	Mean	Median	t-stat	z-stat
Firm characteristics						
Log(S)	4.77	4.80	4.82	4.81	-1.44**	-22.10
Size	0.77	0.80	1.43	1.40	-1.08**	-1.96
Tangibility	0.20	0.20	0.28	0.25	-0.06**	-4.21
ROE	2.48	1.83	2.47	1.80	-1.00	-1.00
Operating CF	0.16	0.13	0.15	0.14	-0.01	-1.70
Leverage (Debt)	0.16	0.13	0.15	0.14	-0.01	-1.70
Ratio	0.45	0.40	0.40	0.35	-0.05	-5.75
Term Spread	0.94	0.91	1.15	1.20	-0.23**	-4.85
Credit Spread	0.86	0.86	0.84	0.80	-0.04*	-1.97
Ratio	1.00	1.00	1.00	1.00	-0.01	-0.12
Loan characteristics						
Log(S)	4.81	4.80	4.80	4.80	-0.01	-1.01
% Senior Debt	0.07	0.00	0.00	0.00	0.07	1.01
Transfer CEO	0.00	0.00	0.00	0.00	0.00	0.00
CEO Ownership %	0.02	0.00	0.02	0.00	0.00	0.00
CEO Retirement Age	0.00	0.00	0.00	0.00	0.00	0.00
No. New Appoint	0.00	0.00	0.00	0.00	0.00	0.00
CEO Age	50.10	50.00	50.00	50.00	0.10	1.00
CEO Tenure	6.10	6.00	6.00	6.00	0.10	1.00
No. New CEO	0.00	0.00	0.00	0.00	0.00	0.00

Table I - Continued

Panel B: Univariate Analysis							
	Loan Contract with a CMR		Loan Contract with no CMR		Difference in Means		Witness Rank-Sum Test
	Mean	Median	Mean	Median	Difference	t-stat	
CMR Proxy	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Log(S)	4.77	4.80	4.82	4.81	-1.44**	-22.10	-20.94**
Size	0.77	0.80	1.43	1.40	-1.08**	-1.96	-1.97**
Tangibility	0.20	0.20	0.28	0.25	-0.06**	-4.21	-3.36**
ROE	2.48	1.83	2.47	1.80	-1.00	-1.00	-1.00
Operating CF	0.16	0.13	0.15	0.14	-0.01	-1.70	-1.67**
Leverage (Debt)	0.16	0.13	0.15	0.14	-0.01	-1.70	-1.67**
Ratio	0.45	0.40	0.40	0.35	-0.05	-5.75	-5.75**
Term Spread	0.94	0.91	1.15	1.20	-0.23**	-4.85	-4.42**
Credit Spread	0.86	0.86	0.84	0.80	-0.04*	-1.97	-1.97**
Ratio	1.00	1.00	1.00	1.00	-0.01	-0.12	-0.12**
Loan characteristics							
Log(S)	4.81	4.80	4.80	4.80	-0.01	-1.01	-1.01**
% Senior Debt	0.07	0.00	0.00	0.00	0.07	1.01	1.01**
Transfer CEO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CEO Ownership %	0.02	0.00	0.02	0.00	0.00	0.00	0.00
CEO Retirement Age	0.00	0.00	0.00	0.00	0.00	0.00	0.00
No. New Appoint	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CEO Age	50.10	50.00	50.00	50.00	0.10	1.00	1.00**
CEO Tenure	6.10	6.00	6.00	6.00	0.10	1.00	1.00**
No. New CEO	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Read:

- The sample contains **15,501** loans. The unconditional probability of a CMR clause is **3.4%**.
- CMR contracts are associated with smaller firms, lower Z-scores, lower tangibility, less frequent ratings, and greater CEO-specific human-capital proxies.
- CMR loans also have wider spreads, fewer lenders, shorter maturity, more collateral, and more financial covenants.

Economic message. The raw data already look like banks writing special management protection into loans where default risk and human-capital dependence are high.

5.2 Table II: baseline determinants of CMR inclusion

Read:

- $\log(AT)$ is strongly negative in every specification.
- Z -score is negative and significant.
- Tangibility is negative in most specifications.
- Log number of lenders is negative, while collateral and number of financial covenants are positive in the fuller specifications.
- Log maturity is negative and strongly significant.

Economic message. CMRs show up in contracts where the borrower is fragile and lenders want tighter control. They look complementary to other monitoring tools, not like random legal ornaments.

Table II
Multivariate Analysis-Baseline Specifications

This table reports results from probit regressions. All tests are at the package level unless otherwise specified. The dependent variable is an indicator variable for CMR clause inclusion. We control for credit and term spreads in Specifications (3)–(5). Industry fixed effects are defined using the Fama-French 12-industry classification. Robust standard errors are clustered at the firm level and z -statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Dependent Variable = CMR Clause				
	(1)	(2)	(3)	(4)	(5)
Log(AT)	−0.302*** (−14.39)	−0.316*** (−14.19)	−0.308*** (−11.52)	−0.268*** (−8.74)	−0.232*** (−6.58)
Z-score	−0.040*** (−2.63)	−0.048*** (−2.73)	−0.055** (−2.34)	−0.054** (−2.33)	−0.052** (−2.21)
Tangibility		−0.445* (−1.82)	−0.498** (−1.97)	−0.510** (−2.02)	−0.418 (−1.62)
MtB		−0.002 (−0.28)	−0.004 (−0.42)	−0.003 (−0.38)	−0.002 (−0.22)
Operating CF			0.170 (0.74)	0.214 (0.94)	0.246 (1.04)
Leverage (Book)			0.263 (1.46)	0.283 (1.58)	0.281 (1.54)
Rated			−0.129 (−1.29)	−0.114 (−1.14)	−0.119 (−1.15)
Log (number of lenders)				−0.132** (−2.20)	−0.120* (−1.76)
Scaled loan amount					0.084 (0.69)
Log (Maturity)					−0.193*** (−3.19)
Collateralized					0.146 [†] (1.83)
Number of financial covenants					0.041 [†] (1.71)
Performance pricing					0.012 (0.17)
Loan purpose fixed effects	N	N	N	N	Y
Loan type fixed effects	N	N	N	N	Y
Year fixed effects	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y
Spreads	N	N	Y	Y	Y
Pseudo R^2	0.17	0.17	0.17	0.17	0.19
Observations	14,144	11,827	11,745	11,736	11,237

Table III
CMR Clause Inclusion and Human Capital

This table reports results from probit regressions. The dependent variable is an indicator variable for CMR clause inclusion. The even and odd specifications correspond to Specifications (3) and (5) from Table II, respectively, with the addition of the X variable as listed. We control for credit and term spreads in all specifications. Industry fixed effects are defined using the Fama-French 12-industry classification. Robust standard errors are clustered at the firm level and z -statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

X =	Dependent Variable = CMR Clause					
	Founder CEO		% Insider (Ind.)		No Heir Apparent	
	(1)	(2)	(3)	(4)	(5)	(6)
X	0.392** (2.21)	0.375** (2.00)	0.722** (2.29)	0.653** (2.03)	0.247** (1.96)	0.275** (2.26)
Average marginal effect	0.011	0.010	0.044	0.040	0.006	0.007
Firm controls	Y	Y	Y	Y	Y	Y
Loan/syndicate controls	N	Y	N	Y	N	Y
Loan purpose fixed effects	N	Y	N	Y	N	Y
Loan type fixed effects	N	Y	N	Y	N	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y
Pseudo R^2	0.23	0.27	0.17	0.19	0.22	0.26
Observations	5,818	5,347	11,685	11,180	5,561	5,100

5.3 Table III: human capital proxies

Read:

- **Founder CEO** enters positively and significantly.
- **% Insider (Ind.)** enters positively and significantly.
- **No Heir Apparent** enters positively and significantly.

- The average marginal effects imply economically meaningful increases in CMR likelihood: about **1.0 pp** for founders, **0.52 pp** for a one-standard-deviation increase in inside-promotion industries, and about **0.7 pp** for lack of an heir apparent.

Economic message. Lenders appear to write CMRs when replacing the CEO is hard because her skills are uniquely tied to the firm or because succession planning is weak.

5.4 Table IV: contracting frictions

Table IV
This table reports results from probit regressions. The dependent variable is an indicator variable for CMR clause inclusion. Panel A presents probit regressions of CMR inclusion on firm attributes and contracting frictions. CEO Ownership %, CEO No Unvested Equity, Low NC Enforcement, and CEO Retirement Age. We interact CEO Retirement Age in Panel C by the Heir Apparent variable. Panels B and C decompose the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel D shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel E shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel F shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel G shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel H shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel I shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel J shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel K shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel L shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel M shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel N shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel O shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel P shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel Q shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel R shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel S shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel T shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel U shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel V shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel W shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel X shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel Y shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion. Panel Z shows the effects of founder status (Founder CEO and % Insider Ind.) and the Heir Apparent variable (Heir Apparent and No Heir Apparent) on the probability of CMR inclusion.

Table IV—Continued

Panel B: Human Capital and Contracting Frictions—Nonfounder Retirement

	(1)	(2)	(3)	(4)	(5)
High NC Enforcement × Founder CEO	0.240	0.246			
Low NC Enforcement × Non-Founder CEO	0.135	0.131			
Low NC Enforcement × Founder CEO	0.140	0.145			
Low NC Enforcement × % Insider Ind.	0.468**	0.451**			
High NC Enforcement × % Insider Ind.	0.187	0.182			
Low NC Enforcement × % Insider Ind.	0.181	0.180			
High NC Enforcement × Heir Apparent	0.240	0.235			
Low NC Enforcement × Heir Apparent	0.135	0.131			
Low NC Enforcement × No Heir Apparent	0.140	0.145			
Low NC Enforcement × No Heir Apparent	0.468**	0.451**			
Firm Controls/Year & Industry fixed effects	Y	Y	Y	Y	Y
Fixed Effects & Linear Controls	Y	Y	Y	Y	Y
Pseudo R ²	0.124	0.124	0.124	0.124	0.124
Observations	5,710	5,707	11,039	11,035	5,708

(Continued)

Table IV—Continued

Panel C: Human Capital and Contracting Frictions—CEO Retirement Age

	(1)	(2)	(3)	(4)	(5)	(6)
Not Retirement Age × Founder CEO	0.410**	0.427**				
Retirement Age × Non-Founder CEO	0.135	0.131				
Retirement Age × Founder CEO	0.240	0.246				
Retirement Age × % Insider Ind.	0.468**	0.451**				
Not Retirement Age × % Insider Ind.	0.187	0.182				
Retirement Age × % Insider Ind.	0.181	0.180				
Not Retirement Age × Heir Apparent	0.240	0.235				
Retirement Age × Heir Apparent	0.135	0.131				
Retirement Age × No Heir Apparent	0.140	0.145				
Retirement Age × No Heir Apparent	0.468**	0.451**				
Firm Controls/Year & Industry fixed effects	Y	Y	Y	Y	Y	Y
Fixed Effects & Linear Controls	Y	Y	Y	Y	Y	Y
Pseudo R ²	0.124	0.124	0.124	0.124	0.124	0.124
Observations	5,710	5,707	11,039	11,035	5,708	5,707

(Continued)

Read:

- Higher **CEO Ownership %** is associated with more CMR use.
- **CEO No Unvested Equity** is positive and significant; if shareholders cannot penalize departure through forfeiture, lenders seem more likely to use a CMR.
- **Low NC Enforcement** is positive and significant.
- **CEO Retirement Age** is positive and significant.
- Interactions show that founder status and internal-promotion industries matter even more when noncompetes are weak or retirement risk is present.

Economic message. The covenant is not only about valuable talent. It is especially about valuable talent that shareholders may have trouble retaining on their own.

5.5 Table V: predicted yield differential

Read:

- The coefficient on the predicted yield difference $\log \widehat{Yield}^{NoCMR} - \log \widehat{Yield}^{CMR}$ is **positive and significant** in all three specifications.
- The estimated coefficients are roughly **0.35**, **0.70**, and **0.73** depending on the instrument set.

Economic message. Borrowers are more likely to include CMRs when doing so lowers expected loan yields. That is exactly what efficient risk-sharing predicts and exactly what the collusion story struggles to explain.

Table V CMR Inclusion and Predicted Yield Differential			
This table reports results from probit regressions on the selection-adjusted predicted yield differential and controls. We control for credit and term spreads in all specifications. Industry fixed effects are defined using the Fama-French 12-industry classification. Robust standard errors are clustered at the firm level and z-statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.			
	Dependent Variable = CMR Clause		
	(1)	(2)	(3)
$\widehat{LogYield}_{NoCMR} - \widehat{LogYield}_{CMR}$	0.354*** (3.12)	0.701*** (3.98)	0.725*** (3.79)
Average marginal effect	0.022	0.061	0.069
Firm controls	Y	Y	Y
Loan/syndicate controls	Y	Y	Y
Year fixed effects	Y	Y	Y
Industry fixed effects	Y	Y	Y
Pseudo R^2	0.17	0.16	0.16
Observations	11,237	6,080	4,562

Table VI
The Presence of a CMR Clause and CEO Turnover
This table reports results from probit regressions. The dependent variable is an indicator for CEO turnover. Summary statistics for the sample in this test are presented in the Internet Appendix (Table IA.III, Panel C). Robust standard errors are clustered at the firm level and z-statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Dependent Variable = CEO Turnover _t					
	(1)	(2)	(3)	(4)	(5)	(6)
CMR Clause Binding _t	-0.427** (-2.54)	-0.427*** (-2.59)	-0.388** (-2.03)	-0.409** (-2.45)	-0.417** (-2.54)	-0.386** (-2.03)
TSR 1 year _{t-1}	-0.209*** (-6.12)	-0.217*** (-5.60)	-0.217*** (-5.60)			
TSR 3 year _{t-1}				-0.071*** (-4.24)	-0.081*** (-4.50)	-0.081*** (-4.50)
ROA _{t-1}	-0.461*** (-3.14)	-0.467*** (-3.11)	-0.469*** (-3.12)	-0.342** (-2.22)	-0.334** (-2.13)	-0.336** (-2.14)
CEO High Ownership _{t-1}	-0.284*** (-5.50)	-0.278*** (-5.37)	-0.278*** (-5.35)	-0.284*** (-5.47)	-0.280*** (-5.39)	-0.279*** (-5.37)
CEO Retirement Age _{t-1}	0.604*** (15.87)	0.609*** (15.84)	0.609*** (15.84)	0.601*** (15.78)	0.604*** (15.74)	0.604*** (15.74)
CEO Tenure _{t-1}	0.001*** (5.30)	0.001*** (5.16)	0.001*** (5.16)	0.001*** (5.29)	0.001*** (5.15)	0.001*** (5.15)
CMR Firm			-0.040 (-0.47)			-0.032 (-0.37)
CMR clause binding:						
Average marginal effect	-0.053	-0.052	-0.049	-0.051	-0.052	-0.049
Year fixed effects	N	Y	Y	Y	Y	Y
Pseudo R^2	0.04	0.04	0.04	0.04	0.04	0.04
Observations	16,645	16,645	16,645	16,645	16,645	16,645

Table VII
CMR Inclusion and Future Firm Performance
This table reports results from linear regressions of the firm outcomes—Tobin's Q in Panel A and operating cash flows in Panel B, following Nini, Smith, and Sufi (2009). Outcomes are taken as the variable averages over the first two years after contract initiation less the average over the two prior years. Specifications (1) and (2) are the unconditional regressions of the outcomes on *CMR Clause*. Specifications (3)–(10) decompose the effect of *CMR Clause* into the following X indicator variables for subsample analysis: *Low NC Enforcement*, *CEO Retirement Age*, *Z-score Low* (unmodified Z-score is in the bottom quartile), and *Junk Rating* (firm has a credit rating of BBB or below). Odd specifications include the lagged value of the dependent variable. All specifications include year indicator variables. Robust standard errors are clustered at the firm level and t-statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Dependent Variable = Tobin's Q									
	All Firms		Low NC Enforcement		CEO Retirement Age		Z-score Low		Junk Rating
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(10)
CMR Clause	0.009 (0.26)	0.034 (0.91)							
CMR * X = 1			0.028 (0.56)	0.057 (1.10)	0.107*** (2.25)	0.100** (2.03)	0.007 (0.14)	0.022 (0.56)	0.101** (1.65)
CMR * X = 0			-0.016 (-0.30)	0.007 (0.14)	0.013 (0.15)	0.062 (0.76)	-0.017 (-0.35)	0.015 (0.29)	0.020 (0.35)
Lag Dep. Var.	N	Y	N	Y	N	Y	N	Y	N
R^2	0.09	0.10	0.09	0.10	0.11	0.11	0.11	0.11	0.11
Observations	12,798	11,616	12,479	11,321	6,914	6,697	11,901	10,824	5,844

(Continued)

Table VII—Continued									
Panel B: Dependent Variable = Operating CF									
	All Firms		Low NC Enforcement		CEO Retirement Age		Z-score Low		Junk Rating
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(10)
CMR Clause	0.003 (0.36)	0.003 (0.37)							
CMR * X = 1			-0.007 (-0.75)	-0.009 (-0.94)	0.026** (2.47)	0.020* (1.87)	0.040*** (3.12)	0.027** (2.18)	0.018 (1.69)
CMR * X = 0			0.013 (1.24)	0.013 (1.37)	-0.036 (-1.64)	-0.016 (-0.89)	-0.008 (-0.95)	-0.007 (-0.86)	0.023 (1.32)
Lag Dep. Var.	N	Y	N	Y	N	Y	N	Y	N
R^2	0.04	0.07	0.04	0.07	0.06	0.08	0.05	0.08	0.05
Observations	12,397	11,322	12,085	11,033	6,680	6,525	11,453	10,525	5,984

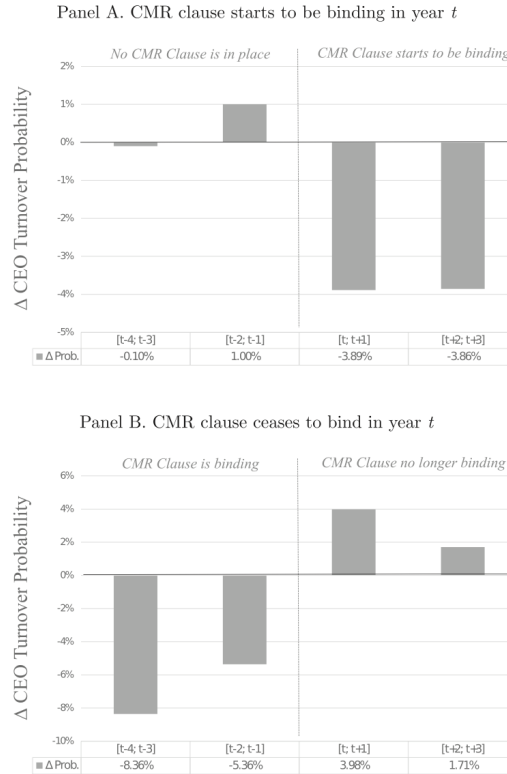


Figure 1. The presence of a CMR clause and abnormal CEO turnover. This figure plots the association of a CMR clause with abnormal CEO turnover. The change in CEO turnover probability is calculated as the difference between realized and predicted turnover, derived from the probit model of Specification (2) in Table VI (excluding *CMR Clause Binding*) on the sample of firms with any contract that contains a CMR. Panel A shows the effect of initiating a debt contract containing a CMR clause at time t . Panel B shows the effect of a cessation of a contract containing a CMR clause at time t .

5.6 Table VI: active CMRs and CEO turnover

Read:

- The coefficient on **CMR Clause Binding** is negative and statistically significant across all specifications.
- The average marginal effect is about **−5.2 percentage points**.
- Given an unconditional turnover rate around **9.49%**, this implies roughly a **55% reduction** in CEO turnover while the clause is active.
- Adding a control for whether the firm ever has a CMR does not eliminate the result.

Economic message. CMRs are not symbolic. They are binding enough to change actual CEO-turnover behavior.

5.7 Figure 1: turnover around CMR initiation and expiration

Read:

- When a firm enters a contract with a CMR, abnormal CEO turnover falls sharply.

- When the CMR contract expires, abnormal turnover rises again.

Economic message. The timing matches the contract, which strongly suggests the clause matters in practice rather than merely tagging certain types of firms.

5.8 Legal evidence: *State National Bank v. Farah Manufacturing Company*

Read:

- After poor performance, William Farah lost the CEO post but stayed on the board.
- When the board considered bringing him back as CEO, lenders threatened to accelerate the loan under the CMR.
- The board instead hired the lender-preferred replacement.

Economic message. This anecdote is not the paper's statistical identification, but it is powerful institutional proof that banks can use these clauses to shape succession outside formal default states.

5.9 Table VII: future Tobin's Q and operating cash flow

Table VII

CMR Inclusion and Future Firm Performance

This table reports results from linear regressions of the firm outcomes—Tobin's Q in Panel A and operating cash flows in Panel B, following Nini, Smith, and Sufi (2009). Outcomes are taken as the variable averages over the first two years after contract initiation less the average over the two prior years. Specifications (1) and (2) are the unconditional regressions of the outcomes on *CMR Clause*. Specifications (3)–(10) decompose the effect of *CMR Clause* into the following X indicator variables for subsample analysis: *Low NC Enforcement*, *CEO Retirement Age*, *Z-score Low* (unmodified Z-score is in the bottom quartile), and *Junk Rating* (firm has a credit rating of BBB or below). Odd specifications include the lagged value of the dependent variable. All specifications include year indicator variables. Robust standard errors are clustered at the firm level and *t*-statistics are reported in parentheses. Variables are defined in the Appendix. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Dependent Variable = Tobin's Q										
	All Firms		Low NC Enforcement		CEO Retirement Age		Z-score Low		Junk Rating	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CMR Clause	0.009 (0.26)	0.034 (0.91)								
CMR * X = 1			0.028 (0.56)	0.057 (1.10)	0.107** (2.25)	0.100** (2.03)	0.007 (0.14)	0.022 (0.56)	0.050 (1.05)	0.101** (2.16)
CMR * X = 0			−0.016 (−0.30)	0.007 (0.14)	0.013 (0.15)	0.062 (0.76)	−0.017 (−0.35)	0.015 (0.29)	0.020 (0.35)	0.003 (0.03)
Lag Dep. Var.	N	Y	N	Y	N	Y	N	Y	N	Y
R ²	0.09	0.10	0.09	0.10	0.11	0.11	0.11	0.11	0.11	0.11
Observations	12,798	11,616	12,479	11,321	6,914	6,697	11,901	10,824	6,123	5,844

(Continued)

Table VII—Continued

Panel B: Dependent Variable = Operating CF

	All Firms		Low NC Enforcement		CEO Retirement Age		Z-score Low		Junk Rating	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CMR Clause	0.003 (0.36)	0.003 (0.37)								
CMR * X = 1			−0.097 (−0.75)	−0.099 (−0.94)	0.026** (2.47)	0.020* (1.87)	0.040*** (3.12)	0.022** (2.18)	0.018 (1.09)	0.028* (1.73)
CMR * X = 0			0.013 (0.24)	0.013 (1.37)	−0.036 (−1.54)	−0.016 (−0.89)	−0.008 (−0.35)	−0.007 (−0.36)	0.023 (1.32)	0.020 (1.33)
Lag Dep. Var.	N	Y	N	Y	N	Y	N	Y	N	Y
R ²	0.04	0.07	0.04	0.07	0.06	0.08	0.05	0.08	0.05	0.08
Observations	12,397	11,322	12,085	11,033	6,680	6,525	11,453	10,525	5,984	5,749

Read:

- In the full sample, the coefficient on **CMR Clause** is positive but insignificant for both Tobin's Q and operating cash flow.
- For firms with CEOs near **retirement age**, the effect on Tobin's Q is about **0.10** and significant; operating cash flow also improves.
- For firms that are **closer to default** (low Z-score or junk rating), operating performance improves and Tobin's Q is more likely to rise.

Economic message. Retaining the CEO is not always valuable, but it looks valuable exactly where the theory says it should be: when the CEO is hard to keep and losing her would be especially costly for lenders and the firm.

6 Short summary

- CMRs are explicit debt covenants that restrict changes in management, usually changes in the CEO.
- They are rare overall, but much more common in small, risky firms and in firms whose CEOs appear especially valuable or hard to replace.
- CMR use is stronger when shareholder-side retention tools are weak, such as under low non-compete enforcement or near CEO retirement.
- CMR inclusion is associated with a lower predicted loan yield, which argues against a pure CEO-entrenchment story.
- While a CMR is active, CEO turnover falls sharply; in the subsamples where retention should matter most, future value and operating performance improve.

7 Conclusion

7.1 Core conclusion

The paper's central conclusion is that lenders sometimes contract directly on management because managerial human capital matters for debt value. CMRs are not generic governance clauses. They are targeted contractual control rights that become attractive when a CEO departure would create large operating or replacement risk.

7.2 Economic intuition

A bank financing a fragile borrower does not only care about cash-flow levels and covenant ratios. It also cares about whether the person running the firm can be replaced without destroying value. If the CEO is unusually important and shareholders cannot reliably keep her, the lender may want a direct contractual lever over retention and succession. A CMR supplies exactly that lever.

7.3 Takeaway

This paper broadens how to think about debt contracting. Debt contracts do not merely respond to management risk indirectly through pricing, maturity, and collateral. They can also assign lenders ex ante control rights over who gets to run the firm. That is a strong and useful lesson for work on financial contracting, governance, and human capital.